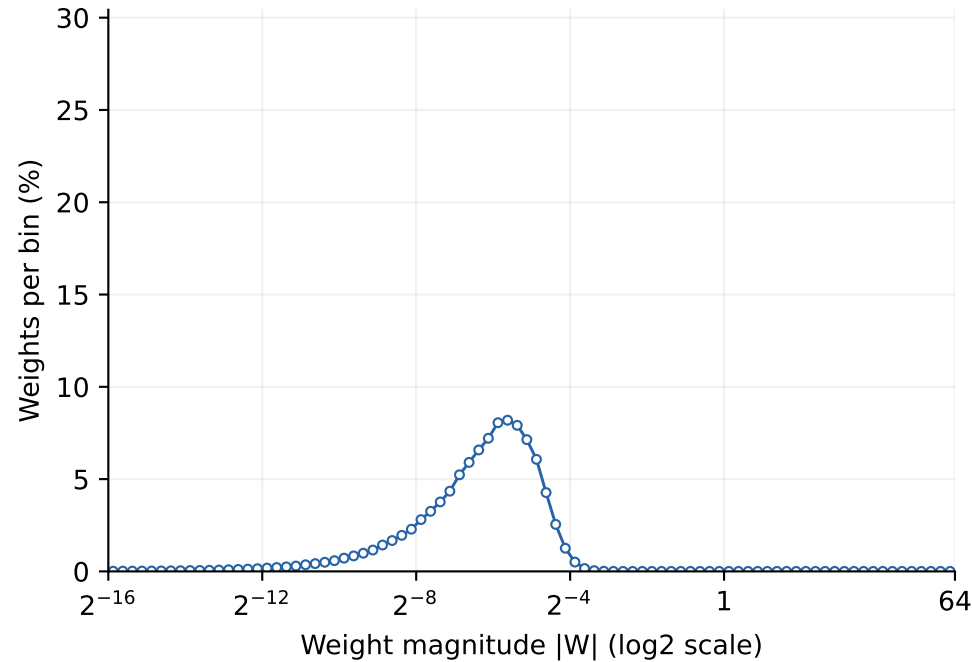


Llama-3.2-1B / Embedding / tied LM head

1 matrix tensor(s) | 262,668,288 weights | FP16 vs PoTaI HP1

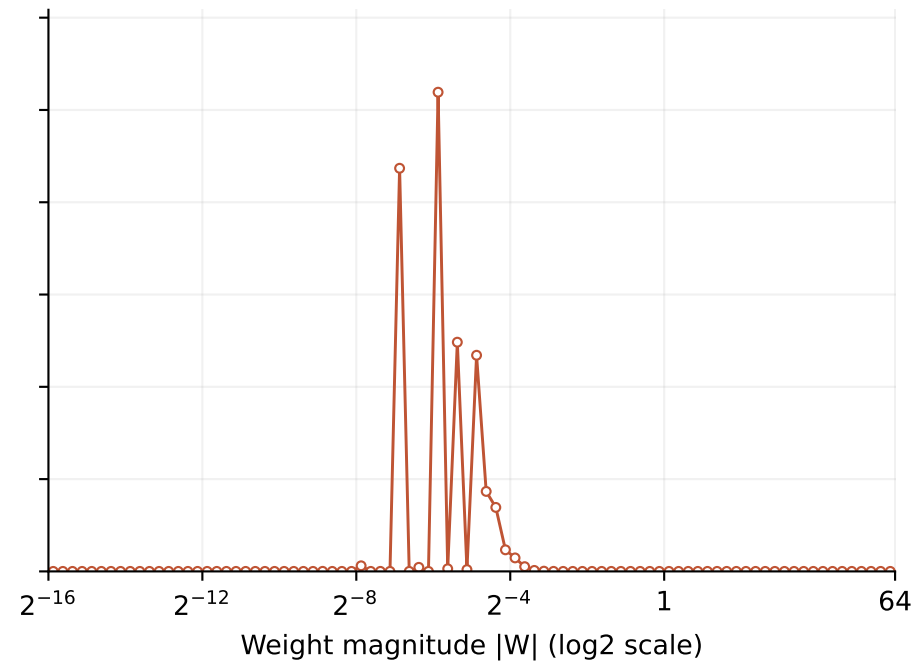
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



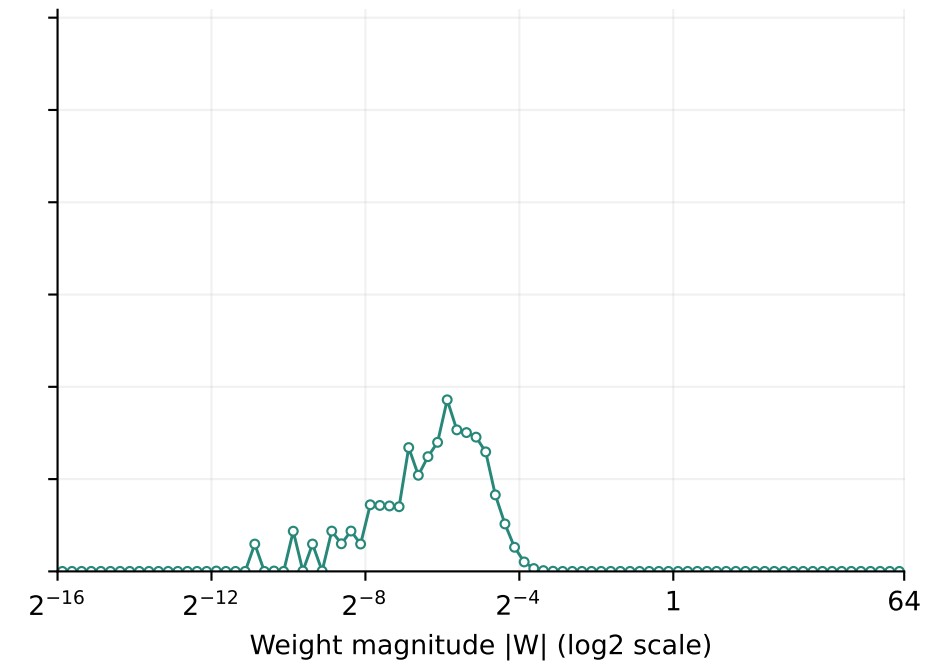
RMS 0.0219068
Max |W| 0.345703
Zero weights 0.0001%
FP16 reference
Nonzero mass outside x-axis 0.0576%

INT4 (Q4_HP1)



RMS 0.0221107
Max |W| 0.375
Zero weights 17.2777%
Relative L2 error 13.035%
SQNR 17.698 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0219691
Max |W| 0.347656
Zero weights 1.0907%
Relative L2 error 0.846%
SQNR 41.452 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: token_embd.

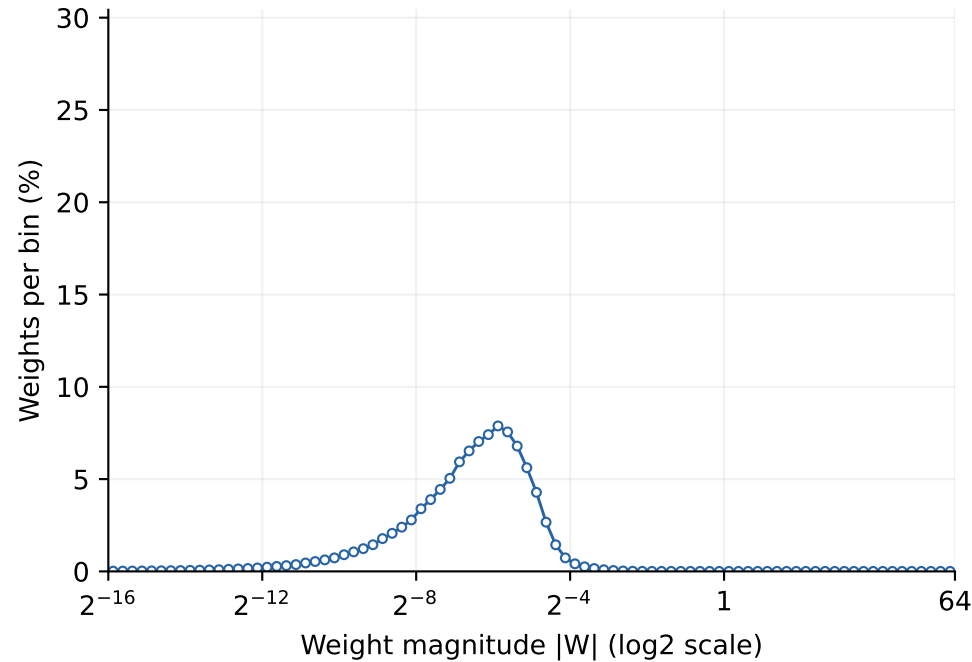
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 00

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

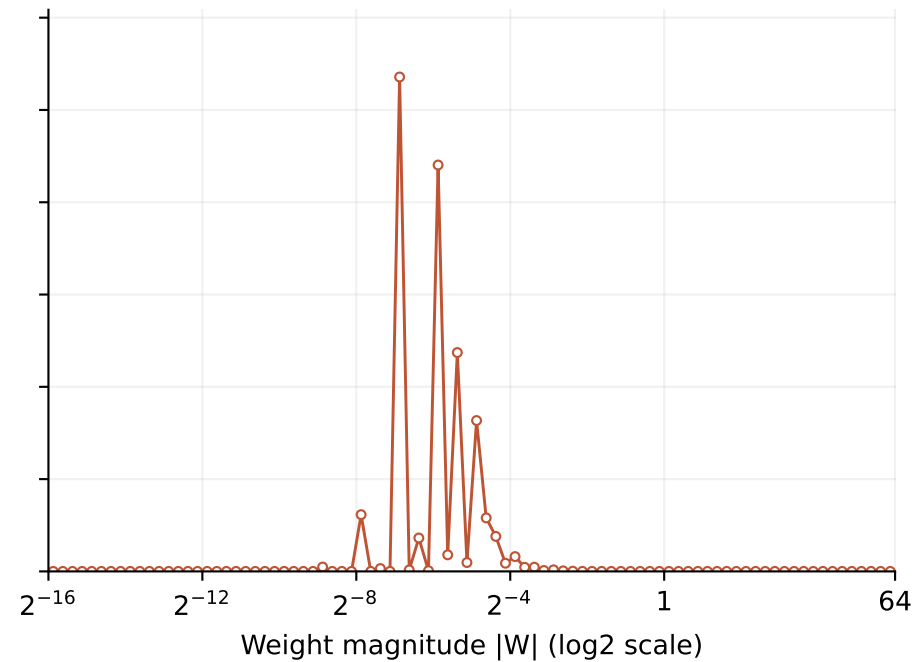
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



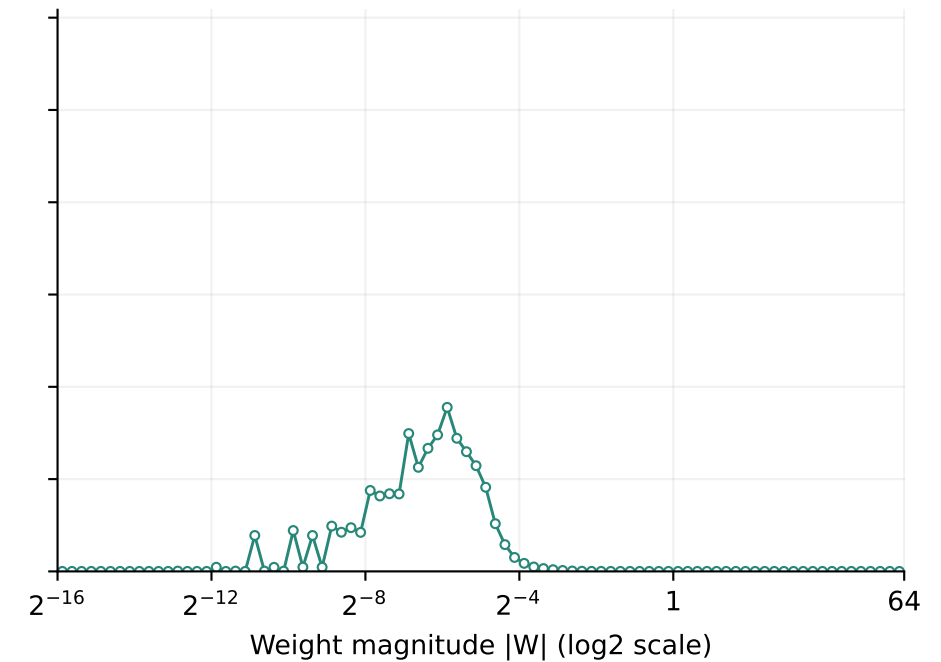
RMS 0.0204517
Max |W| 0.707031
Zero weights 0.0001%
FP16 reference
Nonzero mass outside x-axis 0.0737%

INT4 (Q4_HP1)



RMS 0.0206518
Max |W| 0.75
Zero weights 17.7143%
Relative L2 error 13.034%
SQNR 17.698 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0205101
Max |W| 0.710938
Zero weights 1.1425%
Relative L2 error 0.850%
SQNR 41.413 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

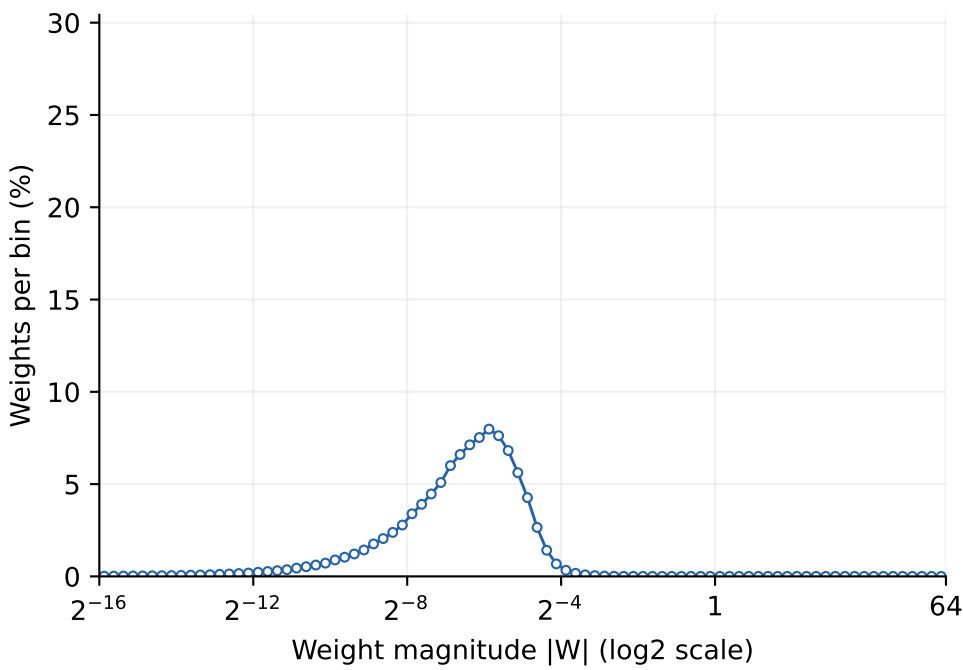
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 01

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

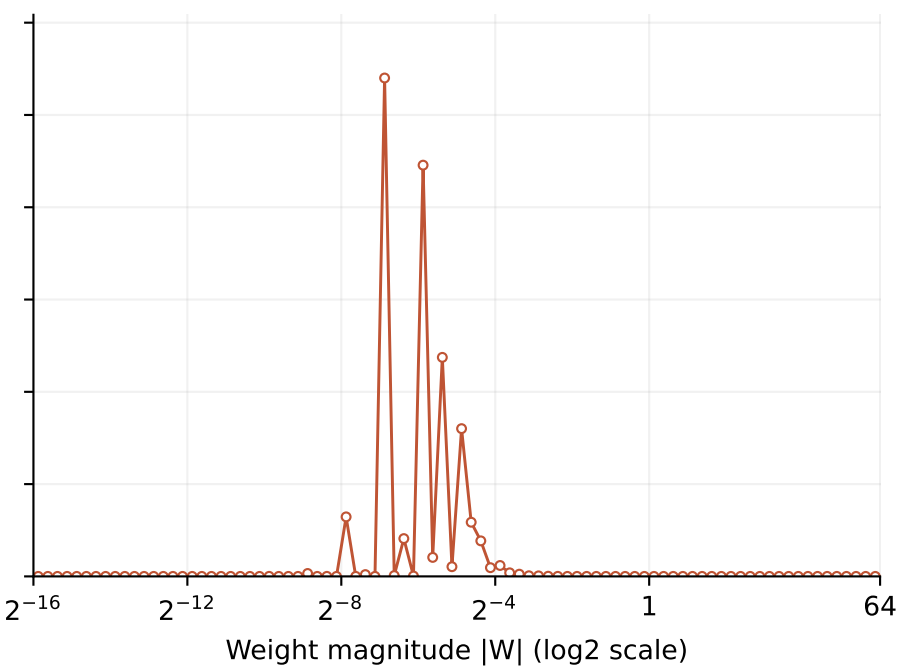
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



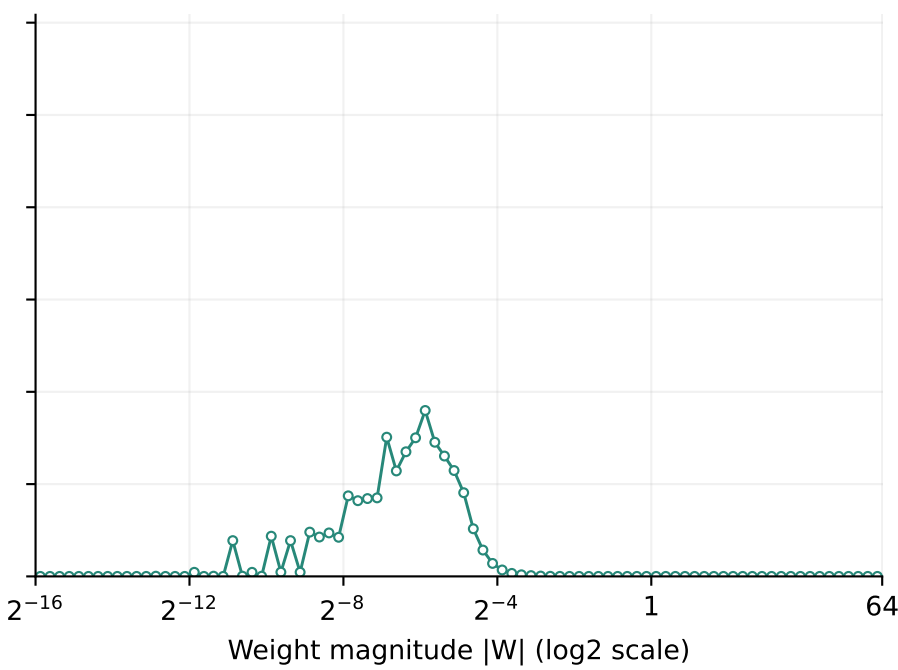
RMS 0.019455
Max |W| 1.125
Zero weights 0.0001%
FP16 reference
Nonzero mass outside x-axis 0.0721%

INT4 (Q4_HP1)



RMS 0.0196432
Max |W| 1.25
Zero weights 17.3775%
Relative L2 error 12.896%
SQNR 17.791 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0195105
Max |W| 1.125
Zero weights 1.1103%
Relative L2 error 0.839%
SQNR 41.524 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

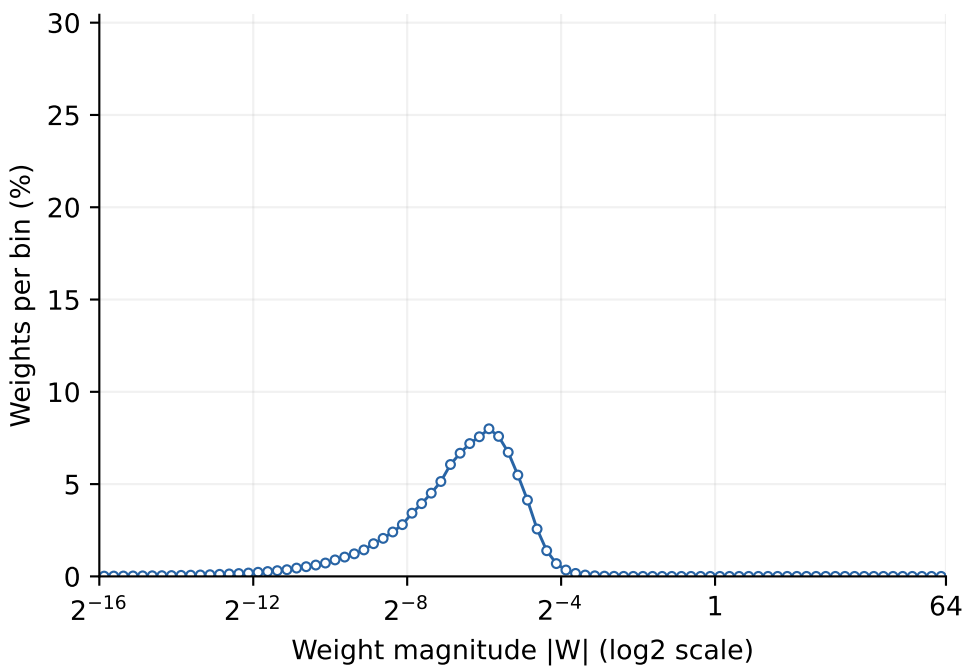
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 02

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

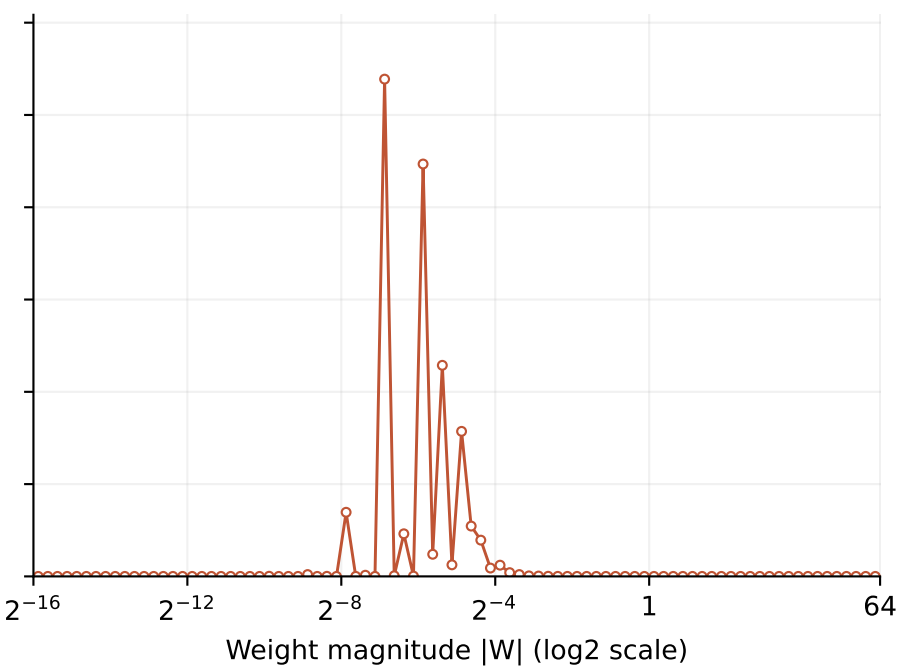
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



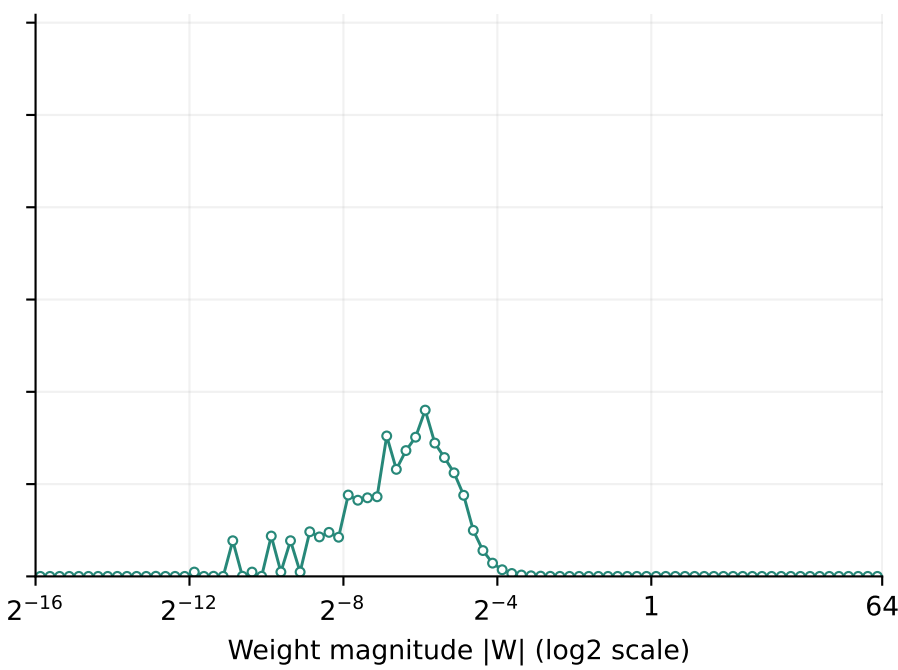
RMS 0.0192286
Max |W| 0.570312
Zero weights 0.0002%
FP16 reference
Nonzero mass outside x-axis 0.0720%

INT4 (Q4_HP1)



RMS 0.0194182
Max |W| 0.625
Zero weights 17.4751%
Relative L2 error 13.070%
SQNR 17.674 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0192834
Max |W| 0.570312
Zero weights 1.1088%
Relative L2 error 0.849%
SQNR 41.425 dB
Nonzero mass outside x-axis 0.0000%

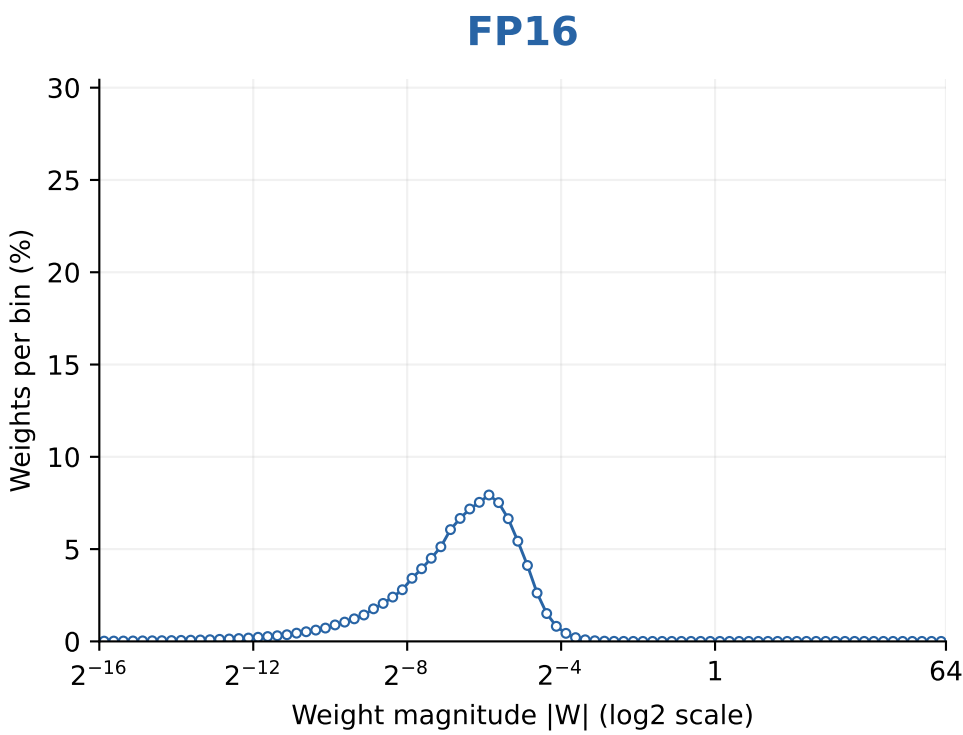
Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

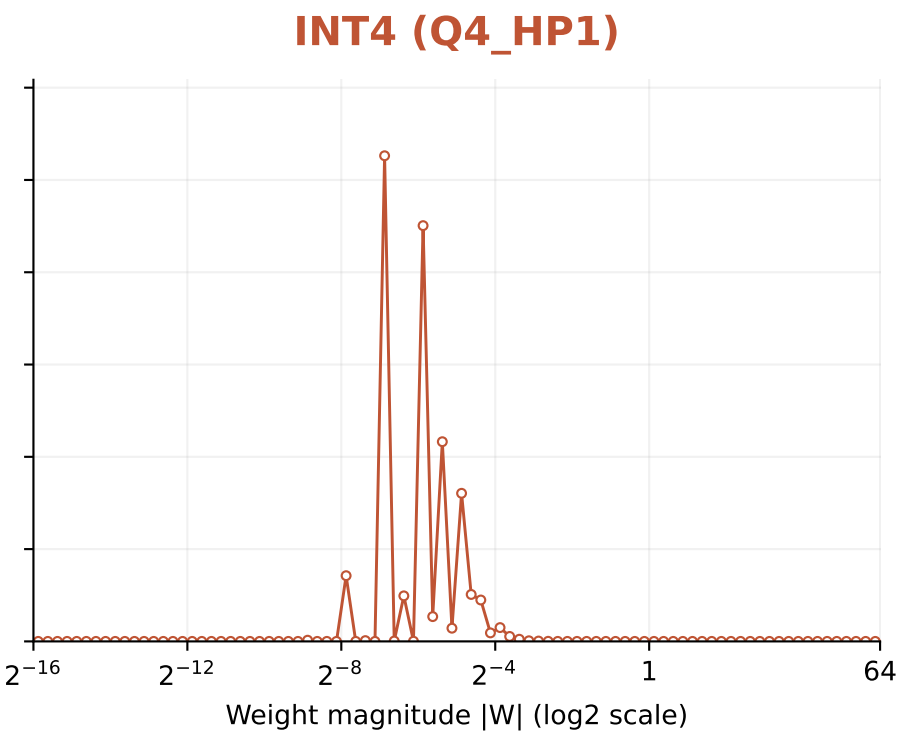
Llama-3.2-1B / Transformer block 03

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

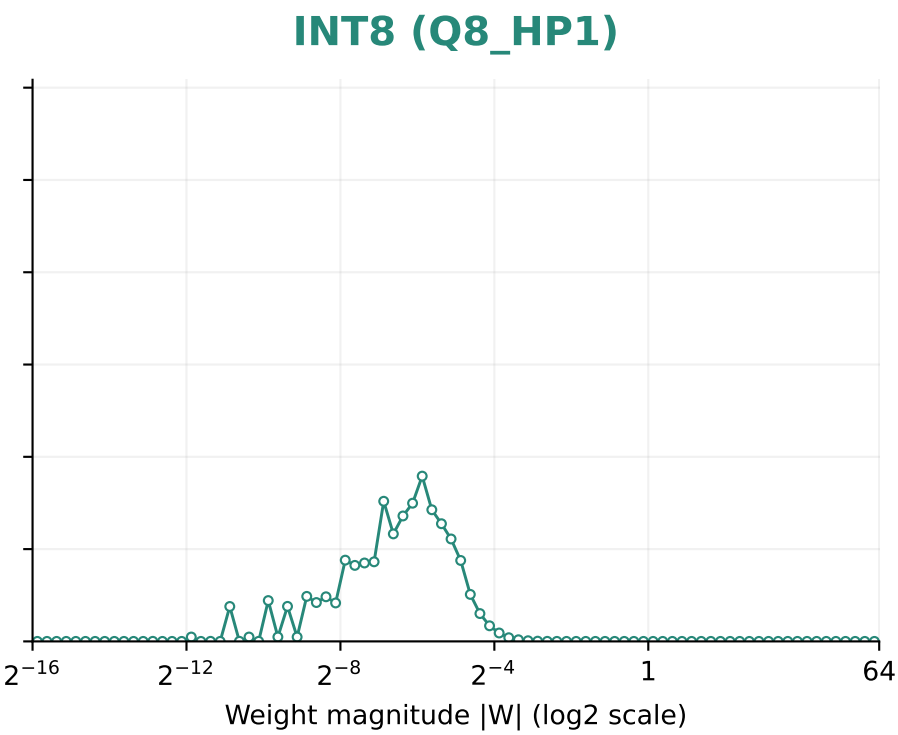
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.



RMS 0.0195551
Max |W| 0.507812
Zero weights 0.0002%
FP16 reference
Nonzero mass outside x-axis 0.0721%



RMS 0.0197467
Max |W| 0.5
Zero weights 17.6585%
Relative L2 error 13.082%
SQNR 17.666 dB
Nonzero mass outside x-axis 0.0000%



RMS 0.0196109
Max |W| 0.507812
Zero weights 1.1197%
Relative L2 error 0.849%
SQNR 41.418 dB
Nonzero mass outside x-axis 0.0000%

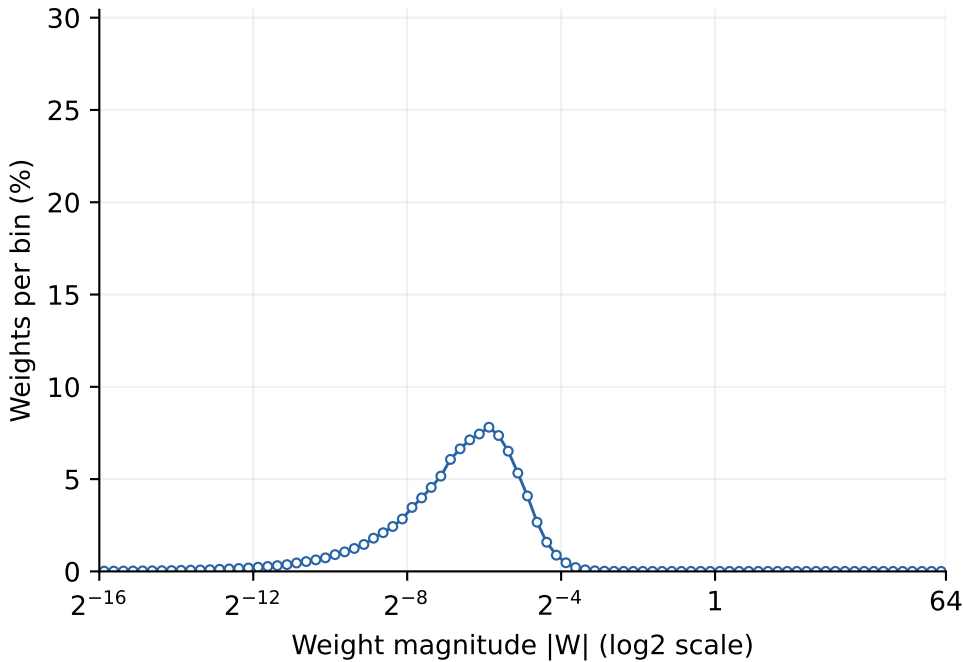
Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 04

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

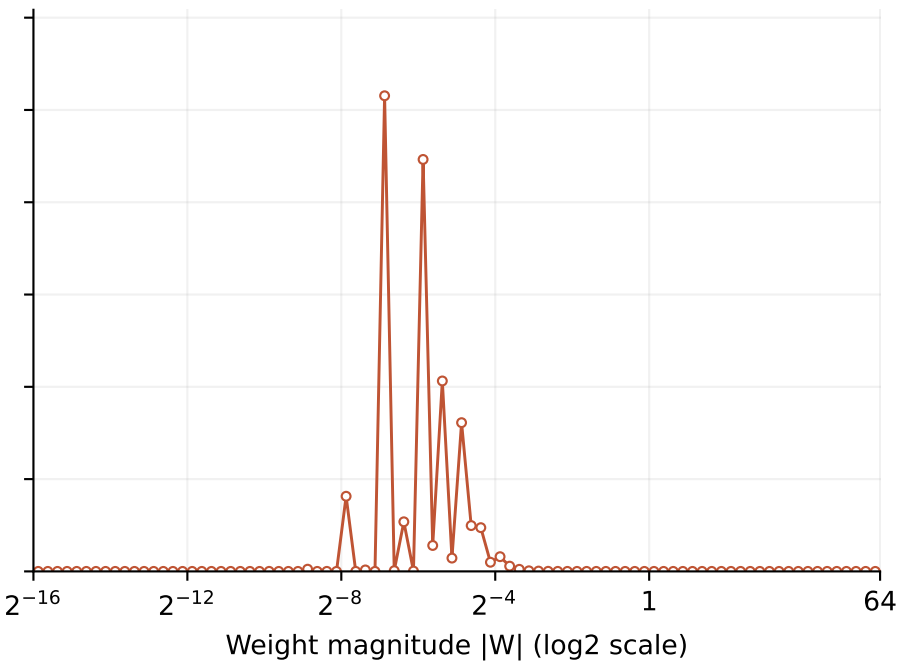
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



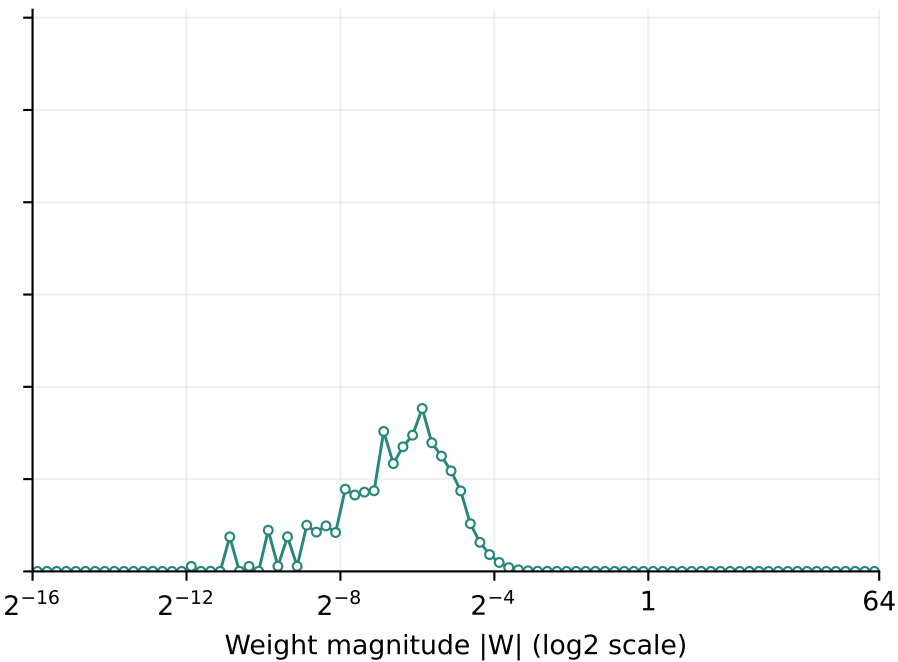
RMS 0.0196389
Max |W| 0.511719
Zero weights 0.0001%
FP16 reference
Nonzero mass outside x-axis 0.0722%

INT4 (Q4_HP1)



RMS 0.0198312
Max |W| 0.5
Zero weights 17.7946%
Relative L2 error 13.184%
SQNR 17.599 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0196948
Max |W| 0.515625
Zero weights 1.1256%
Relative L2 error 0.856%
SQNR 41.353 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

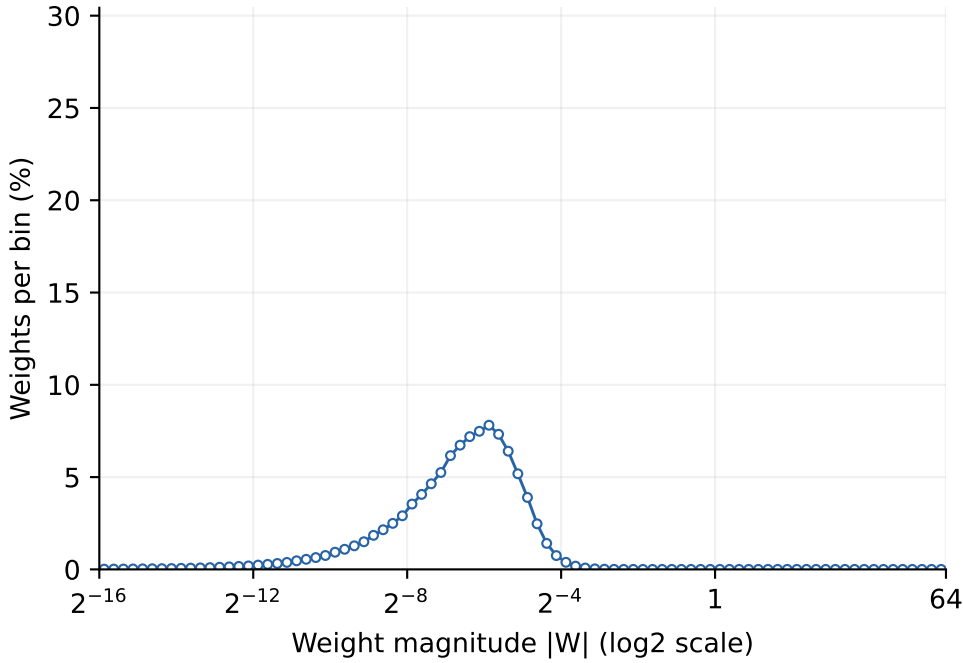
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 05

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

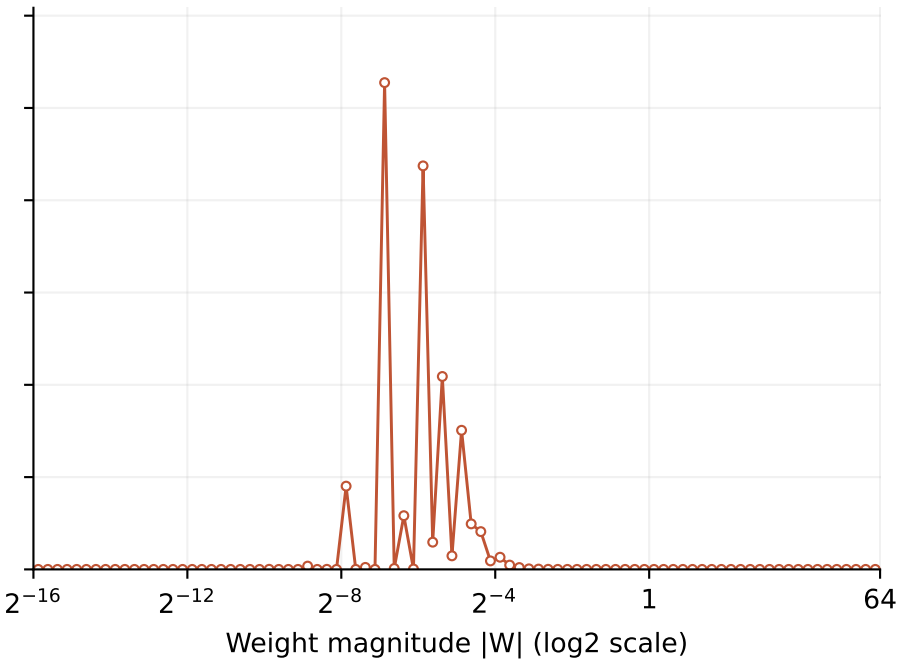
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



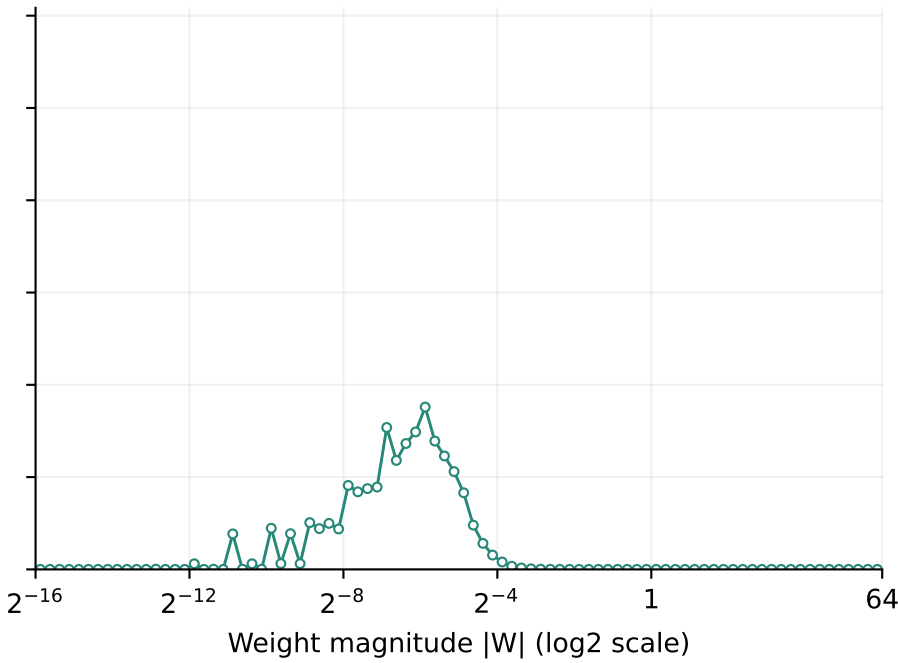
RMS 0.0190671
Max |W| 1.13281
Zero weights 0.0002%
FP16 reference
Nonzero mass outside x-axis 0.0751%

INT4 (Q4_HP1)



RMS 0.019254
Max |W| 1.25
Zero weights 17.7555%
Relative L2 error 13.157%
SQNR 17.617 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0191215
Max |W| 1.14062
Zero weights 1.1257%
Relative L2 error 0.854%
SQNR 41.366 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

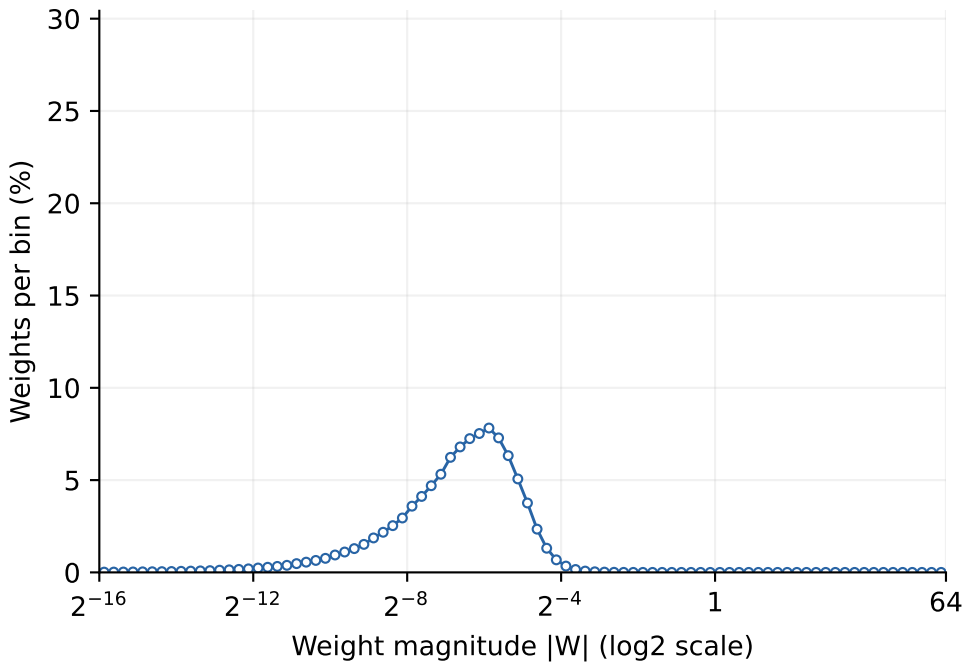
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 06

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

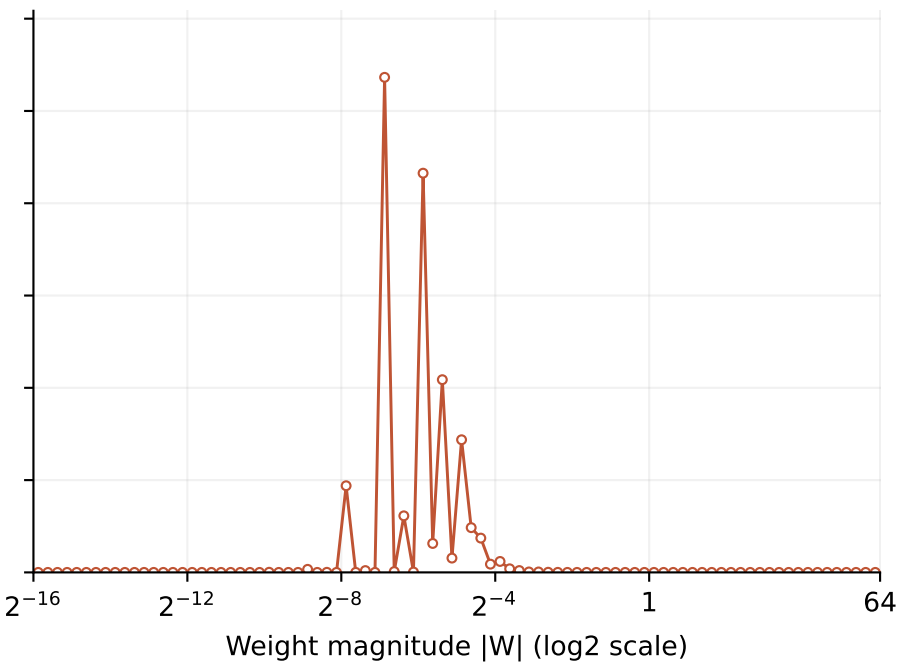
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



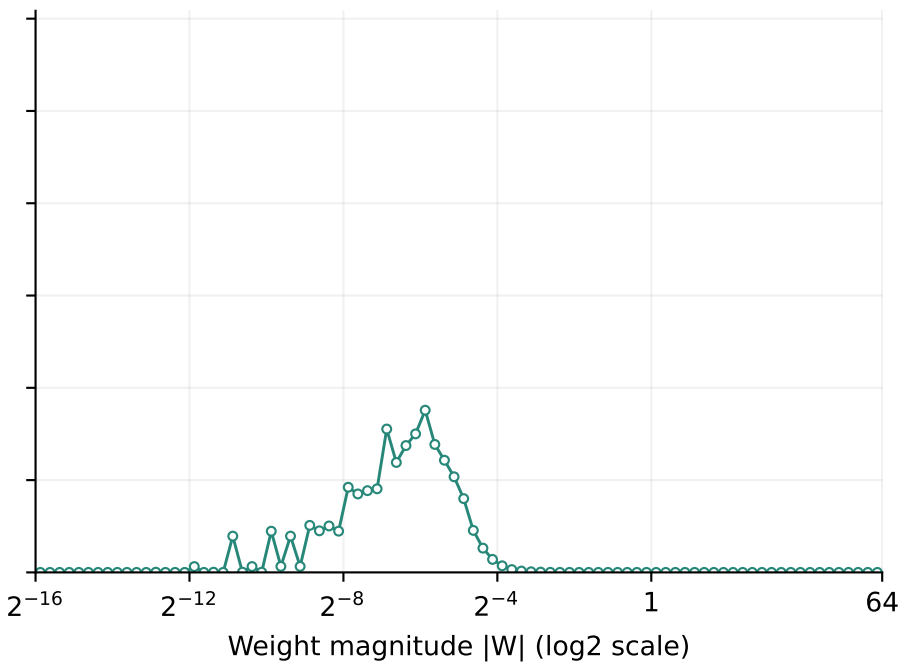
RMS 0.0187674
Max |W| 0.644531
Zero weights 0.0002%
FP16 reference
Nonzero mass outside x-axis 0.0765%

INT4 (Q4_HP1)



RMS 0.0189507
Max |W| 0.625
Zero weights 17.8002%
Relative L2 error 13.149%
SQNR 17.622 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0188209
Max |W| 0.648438
Zero weights 1.1285%
Relative L2 error 0.853%
SQNR 41.377 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

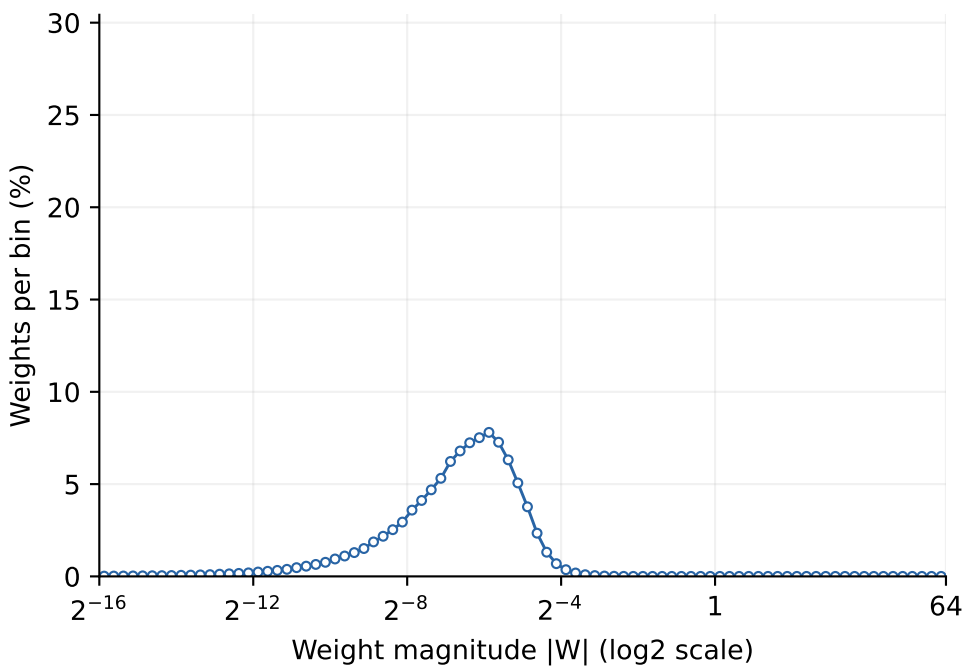
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 07

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

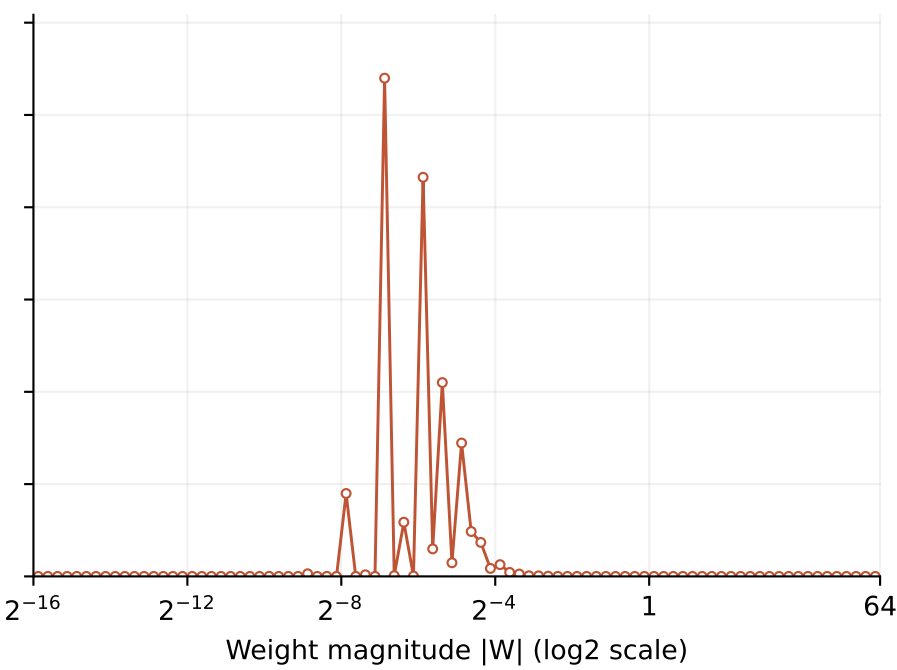
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



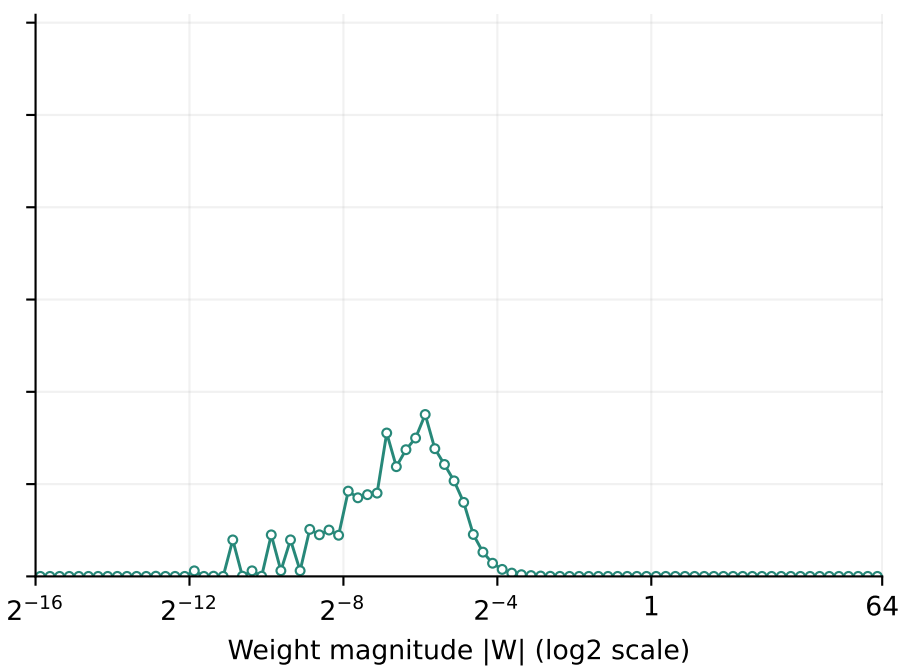
RMS 0.0189571
Max |W| 0.59375
Zero weights 0.0001%
FP16 reference
Nonzero mass outside x-axis 0.0758%

INT4 (Q4_HP1)



RMS 0.0191426
Max |W| 0.625
Zero weights 17.9342%
Relative L2 error 13.149%
SQNR 17.622 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0190112
Max |W| 0.59375
Zero weights 1.1351%
Relative L2 error 0.854%
SQNR 41.372 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

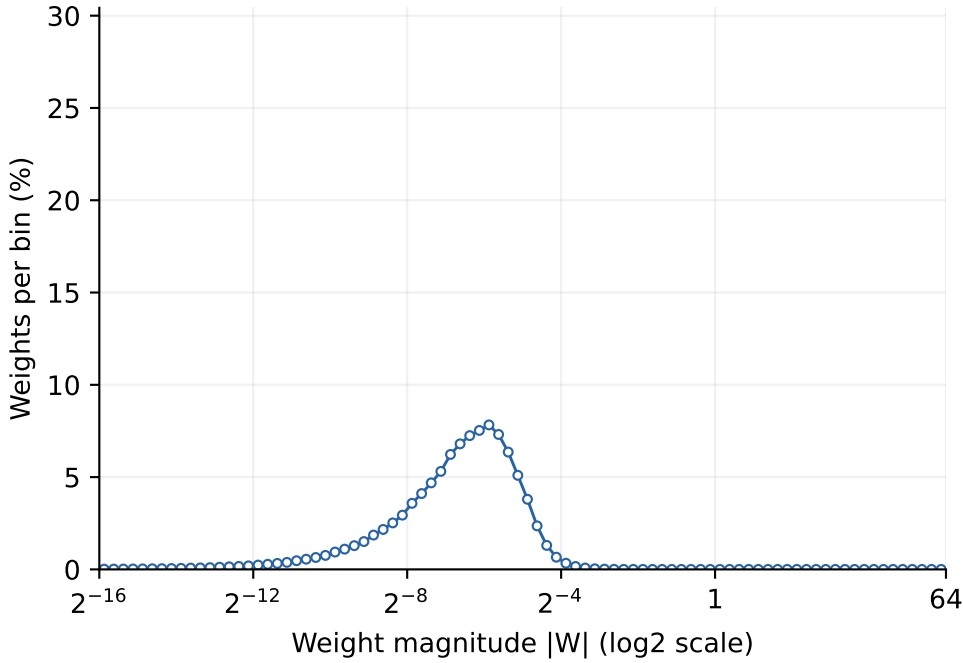
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 08

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

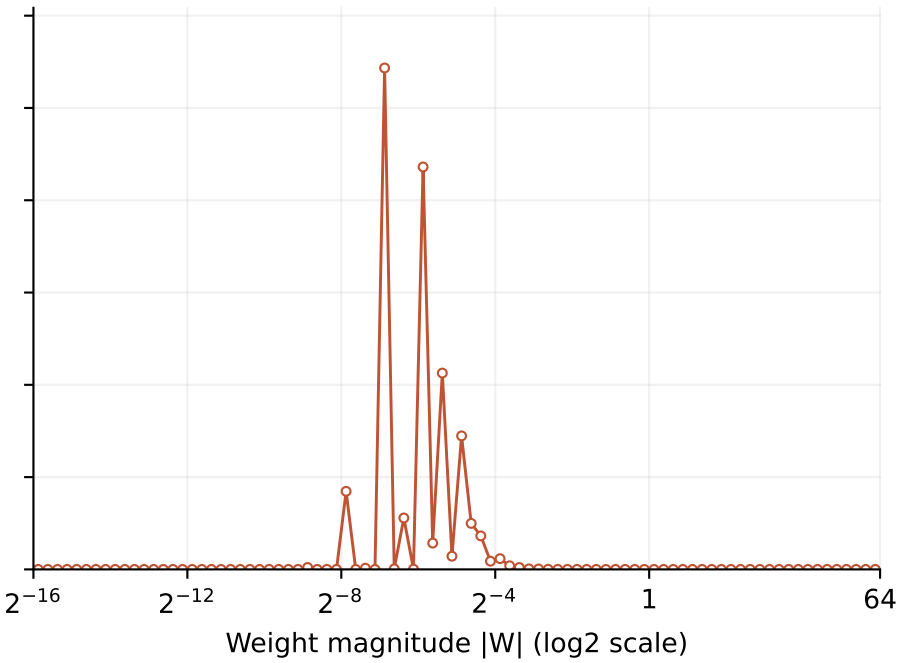
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



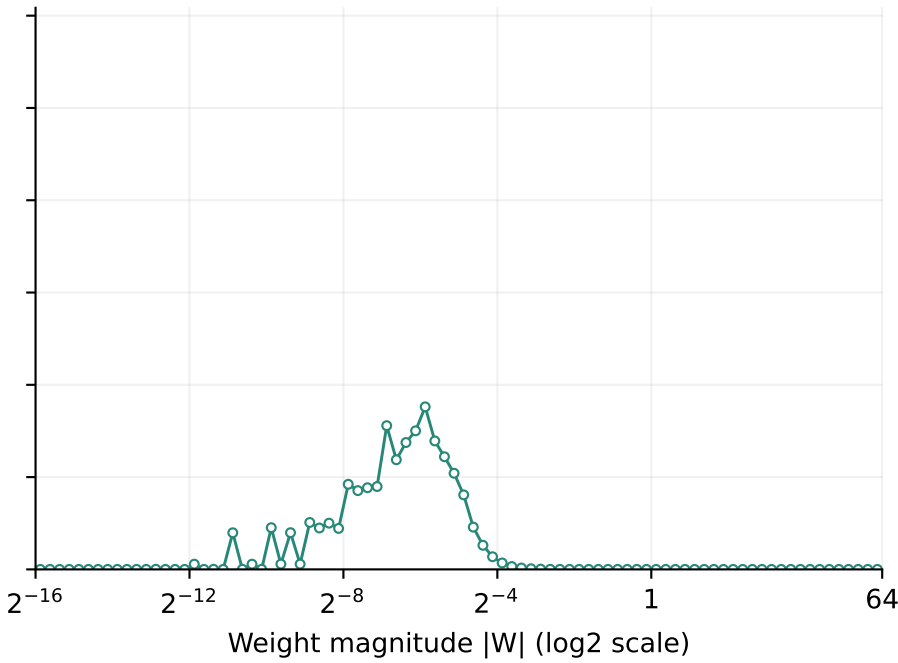
RMS 0.018818
Max |W| 0.582031
Zero weights 0.0002%
FP16 reference
Nonzero mass outside x-axis 0.0753%

INT4 (Q4_HP1)



RMS 0.0190045
Max |W| 0.625
Zero weights 18.0842%
Relative L2 error 13.311%
SQNR 17.516 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0188718
Max |W| 0.585938
Zero weights 1.1474%
Relative L2 error 0.864%
SQNR 41.271 dB
Nonzero mass outside x-axis 0.0000%

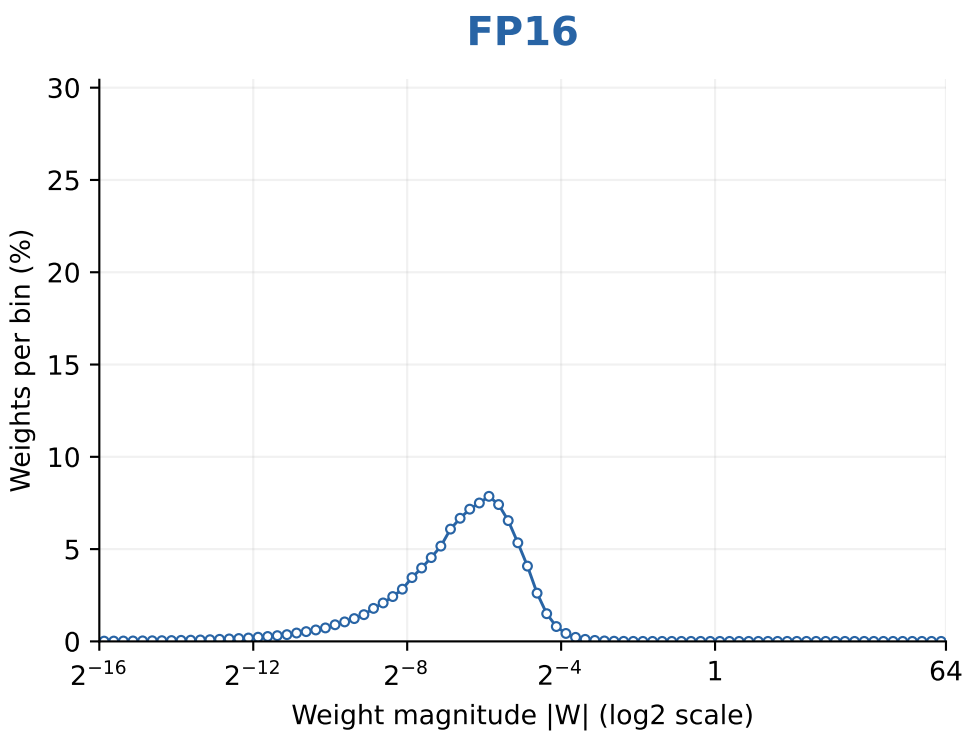
Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

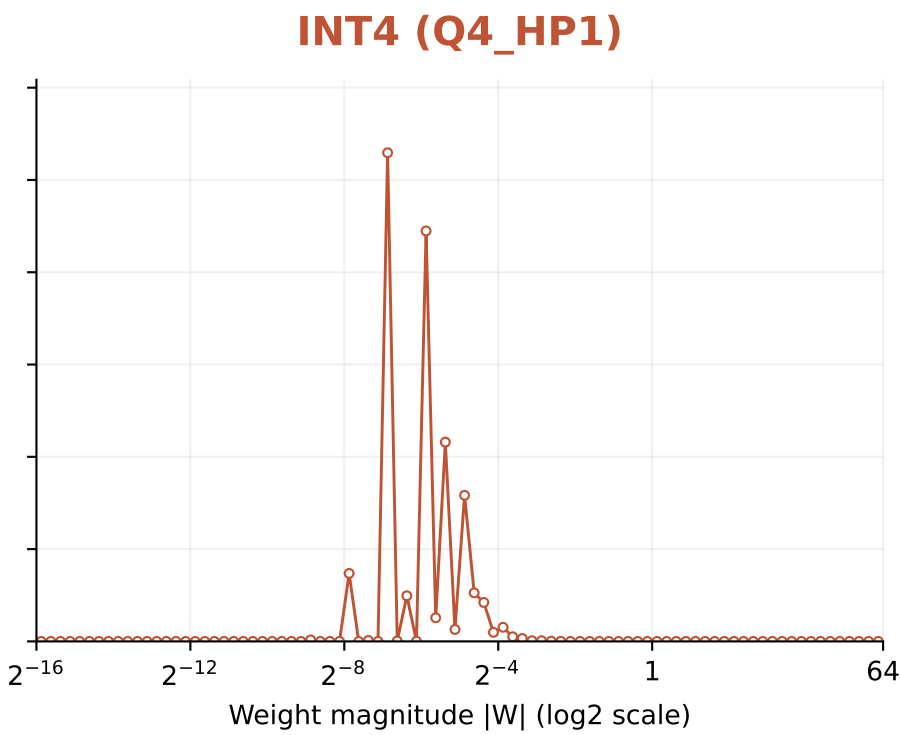
Llama-3.2-1B / Transformer block 09

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

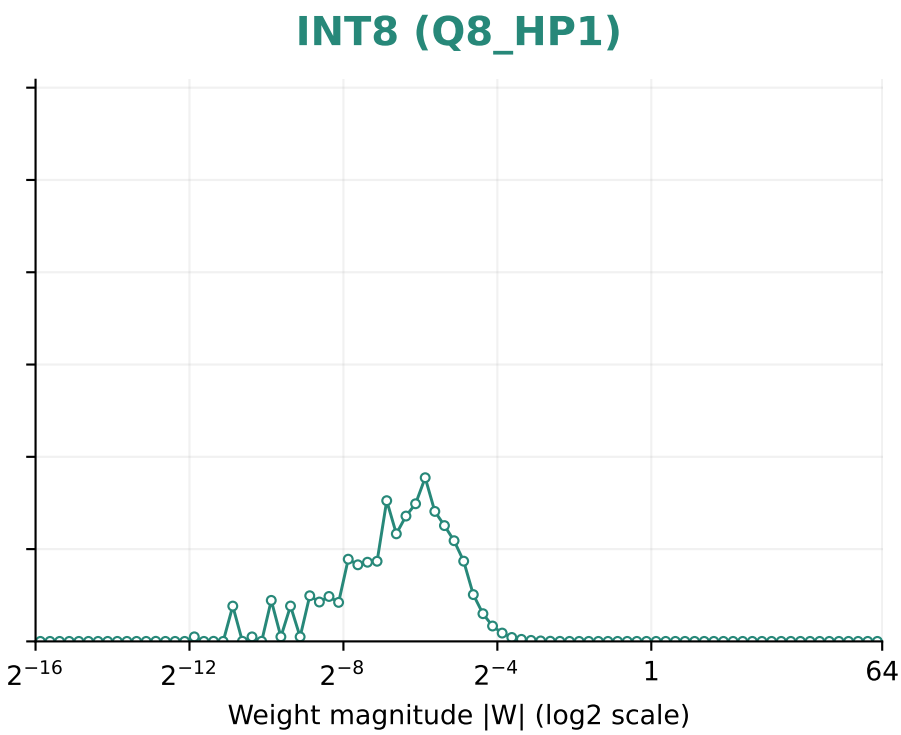
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.



RMS 0.0197603
Max |W| 0.566406
Zero weights 0.0002%
FP16 reference
Nonzero mass outside x-axis 0.0731%



RMS 0.0199536
Max |W| 0.625
Zero weights 17.8160%
Relative L2 error 13.130%
SQNR 17.635 dB
Nonzero mass outside x-axis 0.0000%



RMS 0.0198166
Max |W| 0.570312
Zero weights 1.1288%
Relative L2 error 0.852%
SQNR 41.387 dB
Nonzero mass outside x-axis 0.0000%

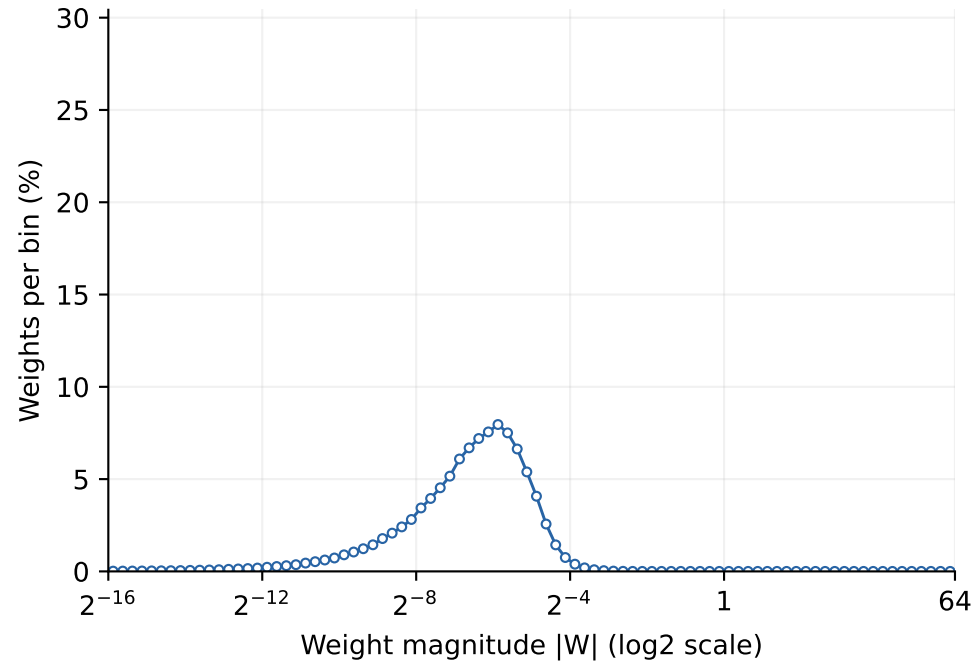
Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 10

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

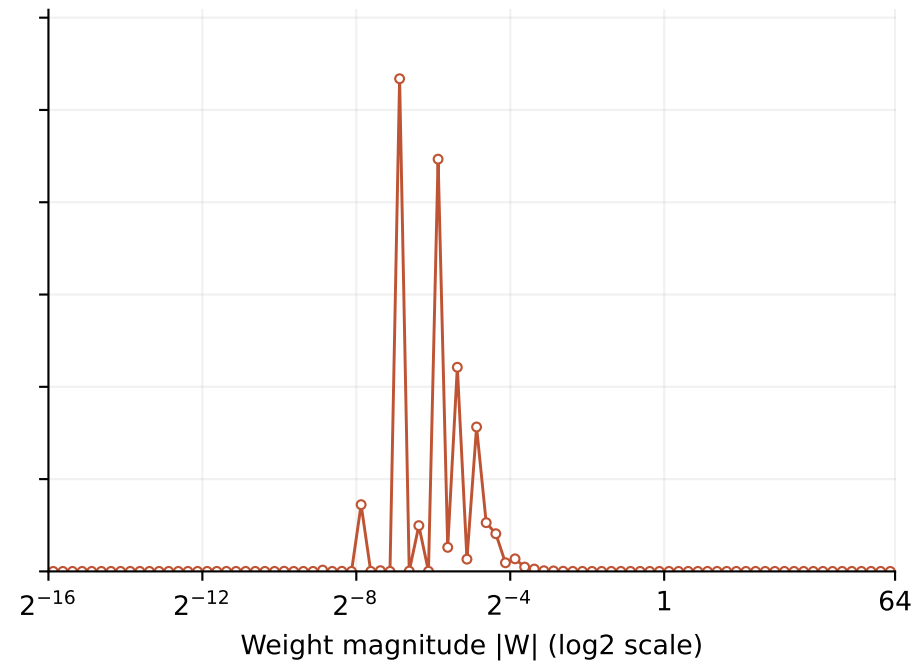
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



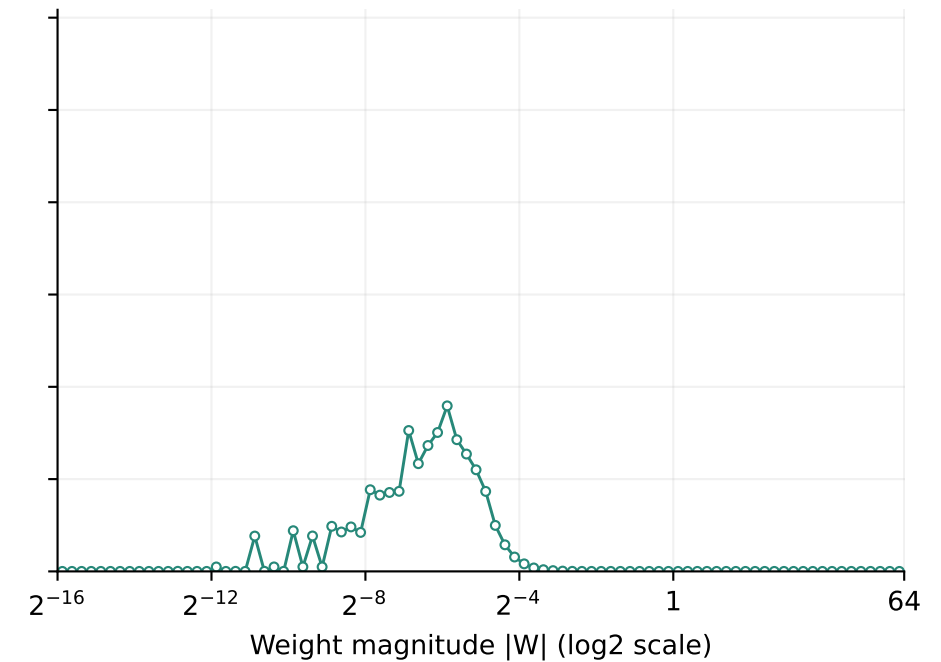
RMS 0.0194203
Max |W| 0.609375
Zero weights 0.0002%
FP16 reference
Nonzero mass outside x-axis 0.0714%

INT4 (Q4_HP1)



RMS 0.0196092
Max |W| 0.625
Zero weights 17.6141%
Relative L2 error 13.052%
SQNR 17.686 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0194756
Max |W| 0.609375
Zero weights 1.1156%
Relative L2 error 0.848%
SQNR 41.434 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

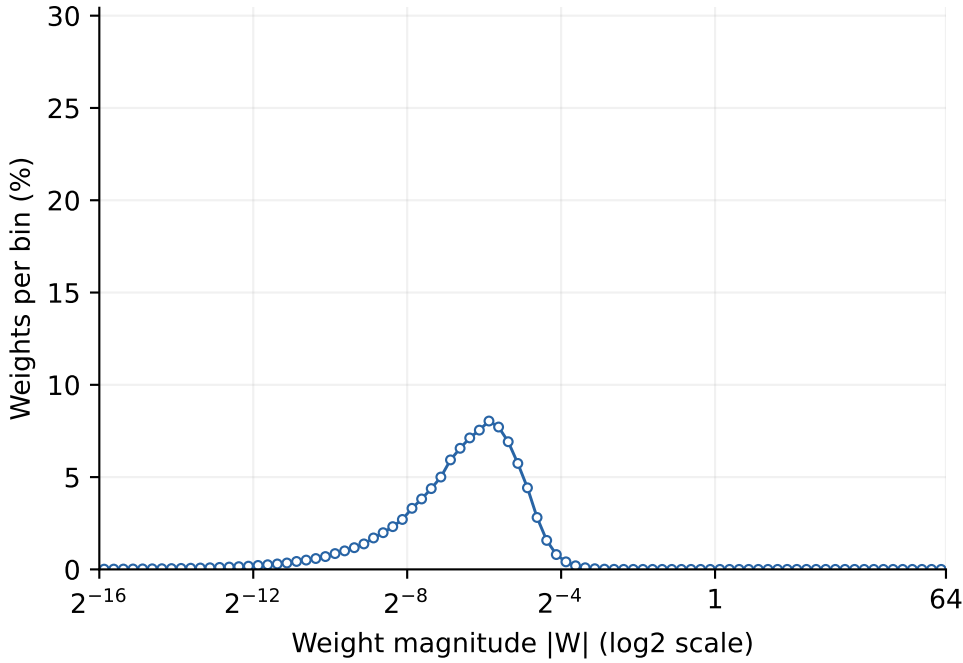
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 11

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

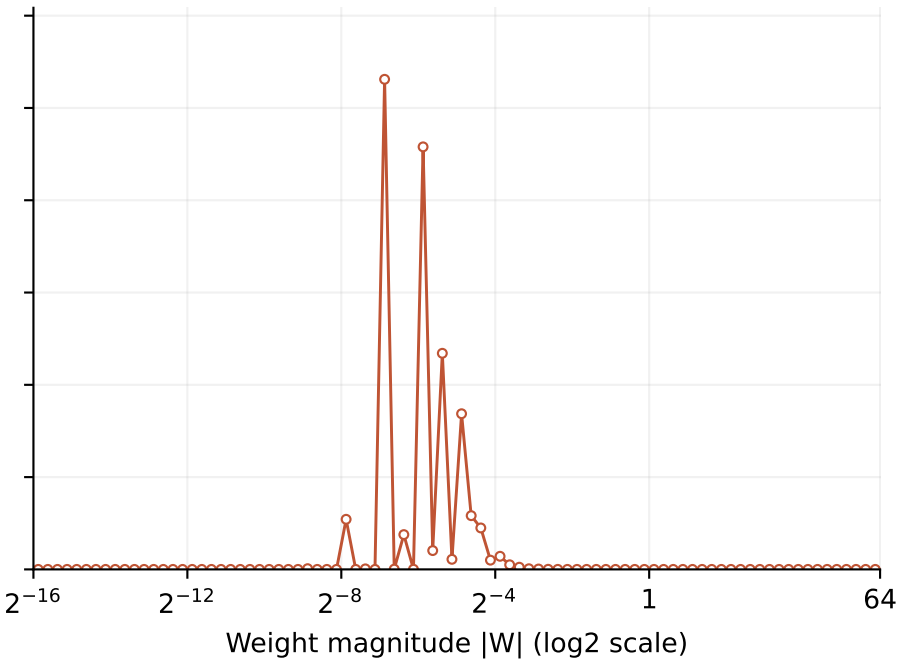
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



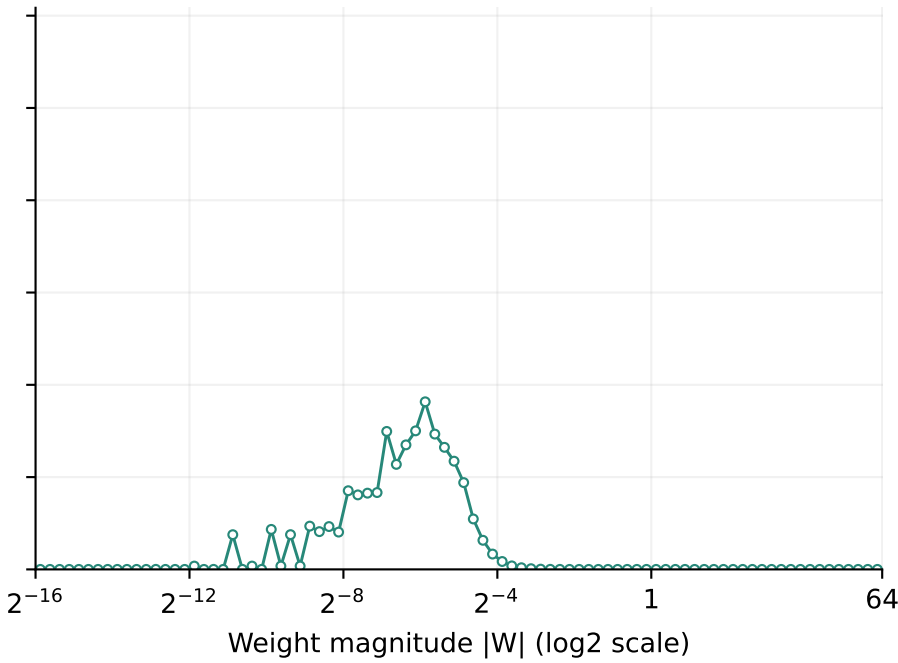
RMS 0.0198629
Max |W| 0.648438
Zero weights 0.0001%
FP16 reference
Nonzero mass outside x-axis 0.0690%

INT4 (Q4_HP1)



RMS 0.0200556
Max |W| 0.625
Zero weights 17.3587%
Relative L2 error 12.915%
SQNR 17.778 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0199195
Max |W| 0.648438
Zero weights 1.0982%
Relative L2 error 0.839%
SQNR 41.520 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

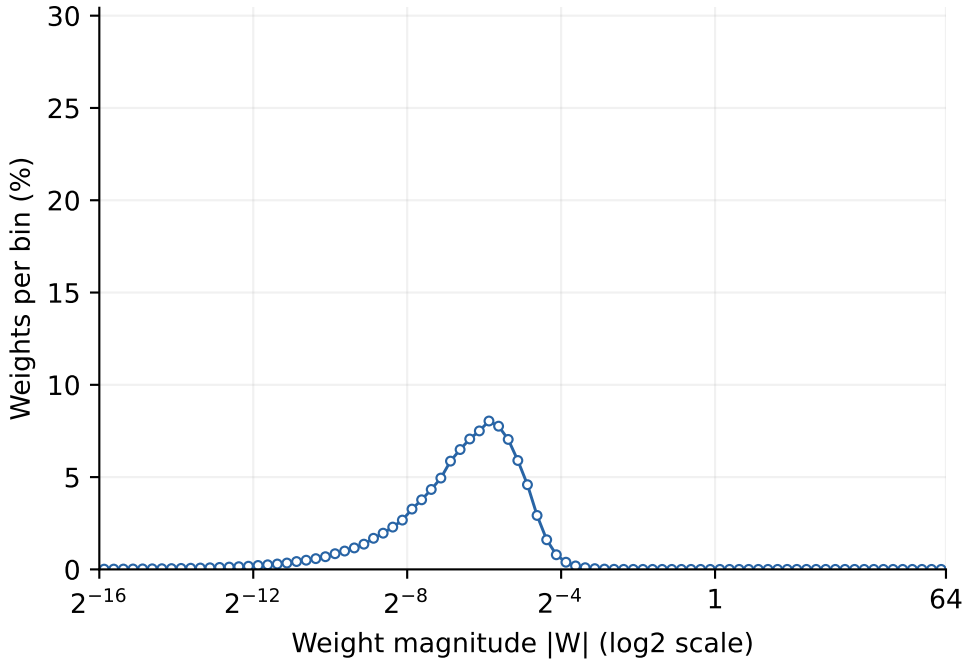
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 12

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

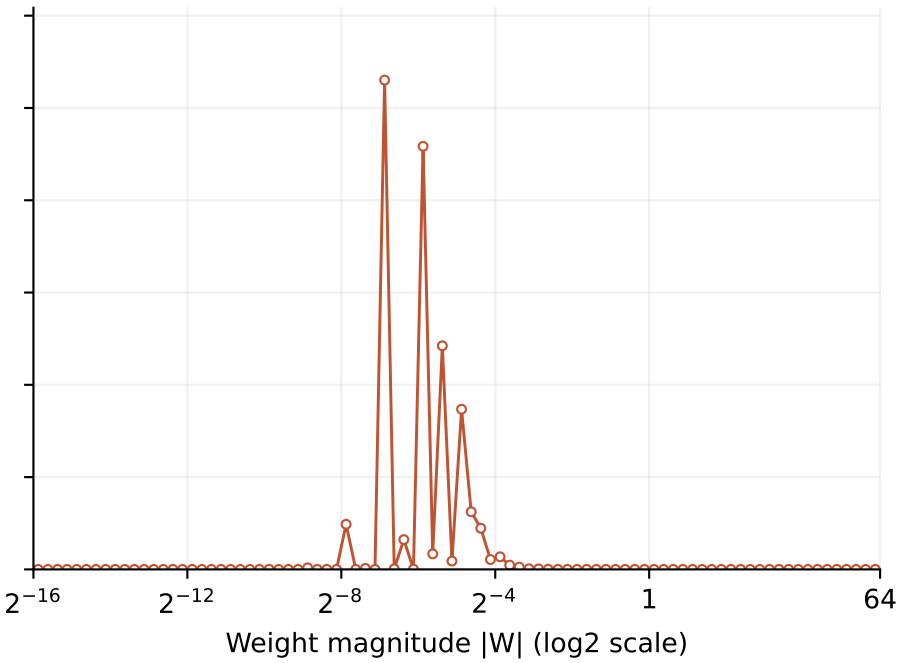
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



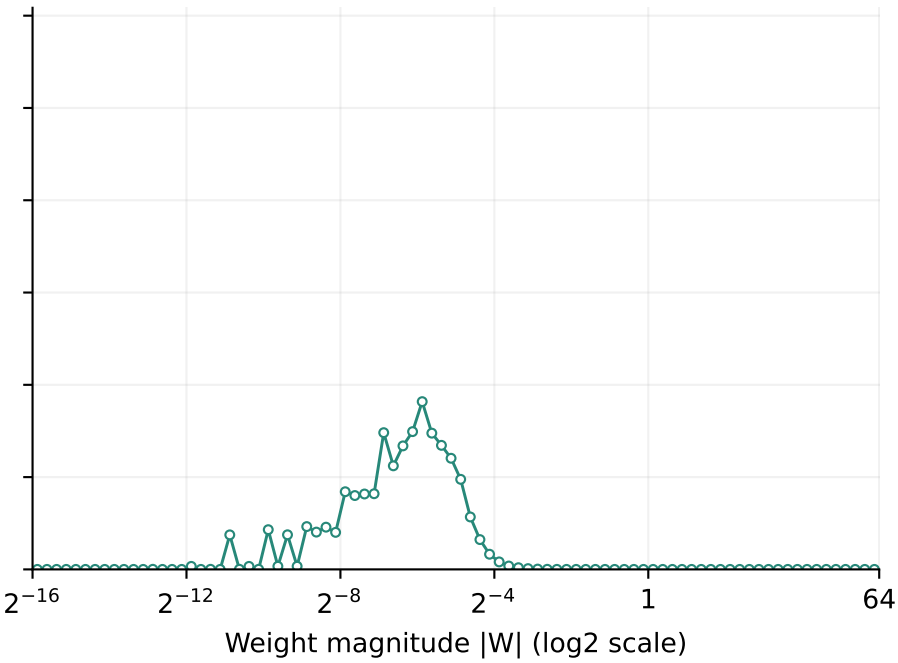
RMS 0.0200327
Max |W| 0.753906
Zero weights 0.0001%
FP16 reference
Nonzero mass outside x-axis 0.0687%

INT4 (Q4_HP1)



RMS 0.0202252
Max |W| 0.75
Zero weights 17.2298%
Relative L2 error 12.872%
SQNR 17.807 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0200898
Max |W| 0.757812
Zero weights 1.0908%
Relative L2 error 0.837%
SQNR 41.543 dB
Nonzero mass outside x-axis 0.0000%

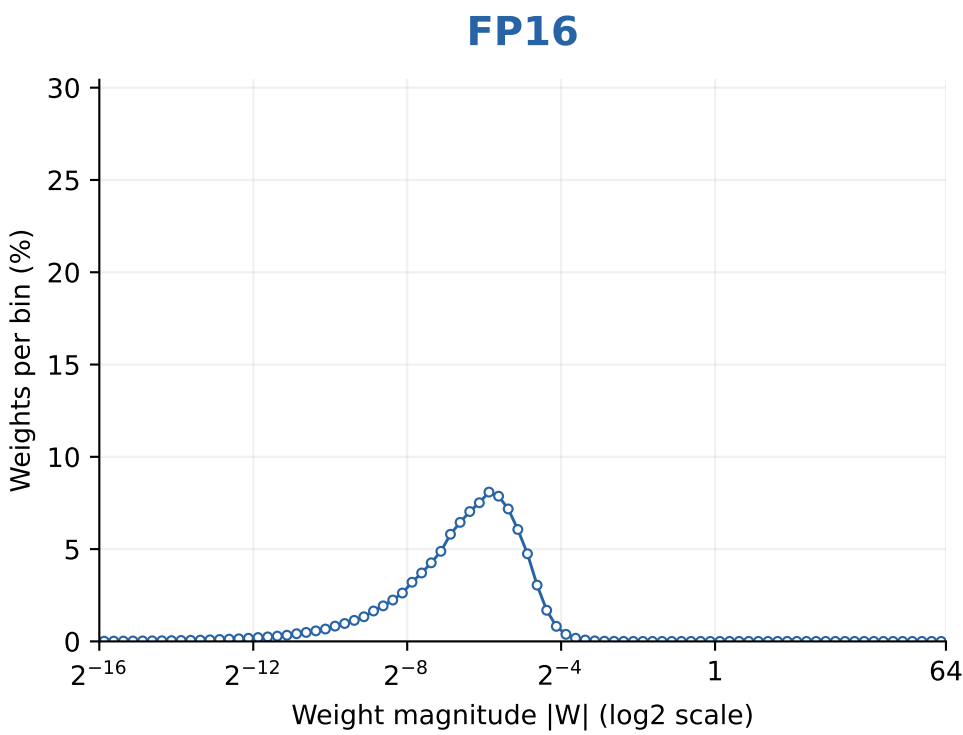
Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

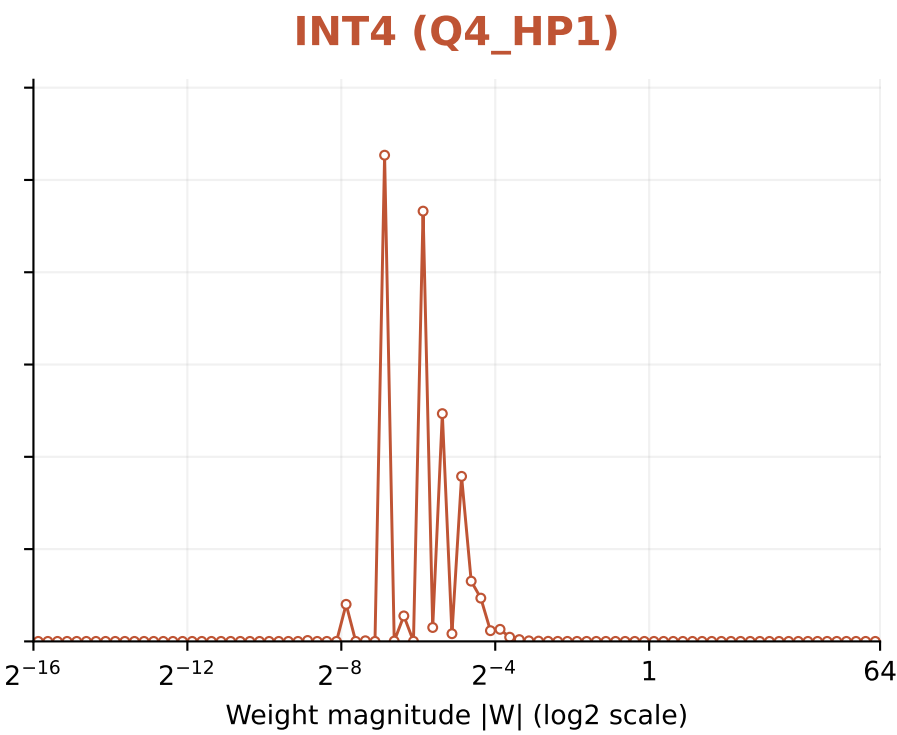
Llama-3.2-1B / Transformer block 13

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

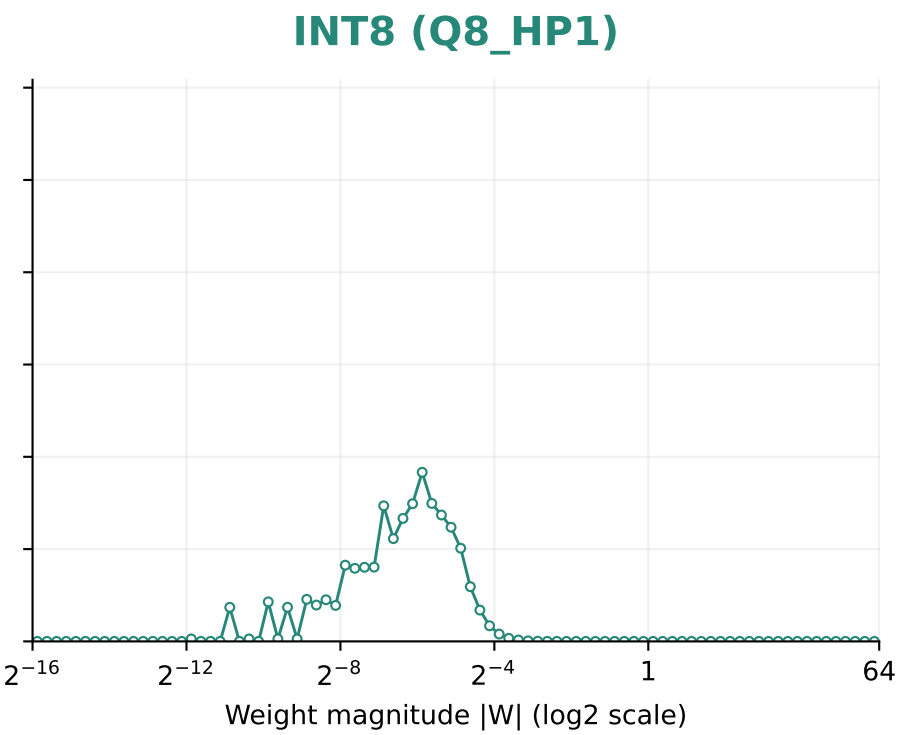
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.



RMS 0.020105
Max |W| 0.667969
Zero weights 0.0001%
FP16 reference
Nonzero mass outside x-axis 0.0671%



RMS 0.0202981
Max |W| 0.625
Zero weights 17.1761%
Relative L2 error 12.850%
SQNR 17.822 dB
Nonzero mass outside x-axis 0.0000%



RMS 0.0201623
Max |W| 0.671875
Zero weights 1.0841%
Relative L2 error 0.836%
SQNR 41.558 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

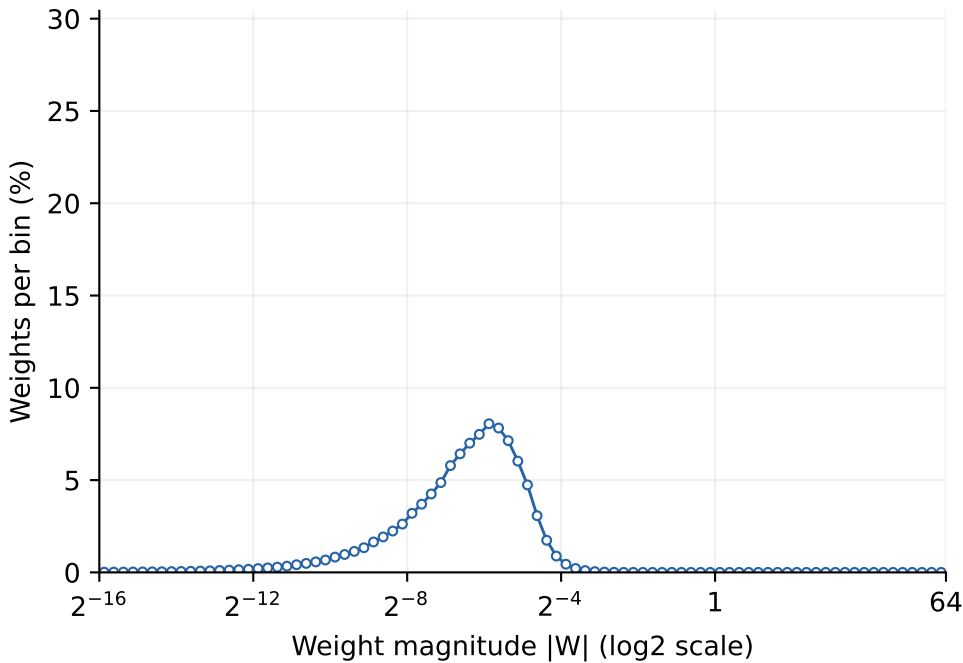
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 14

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

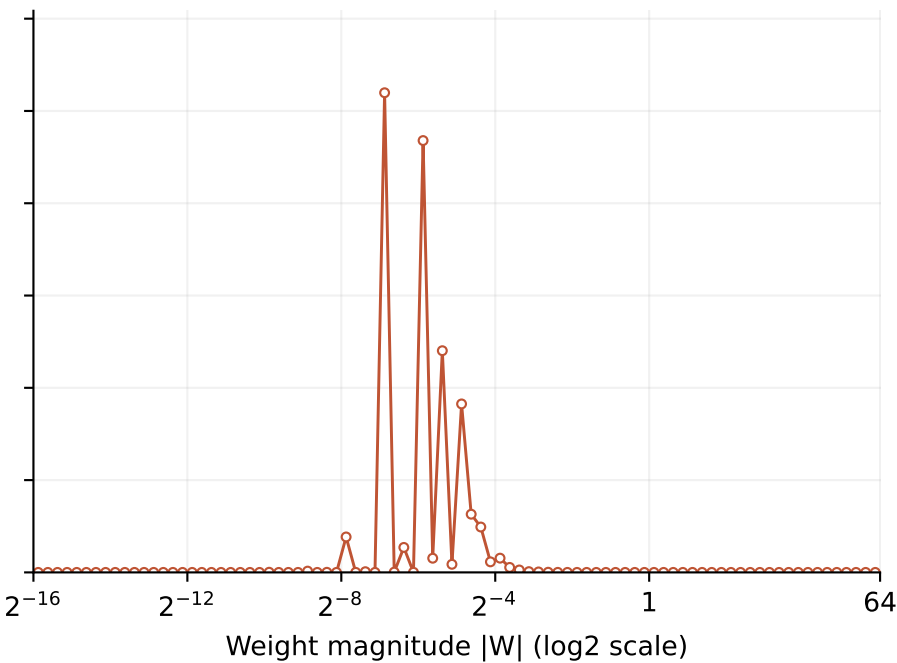
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



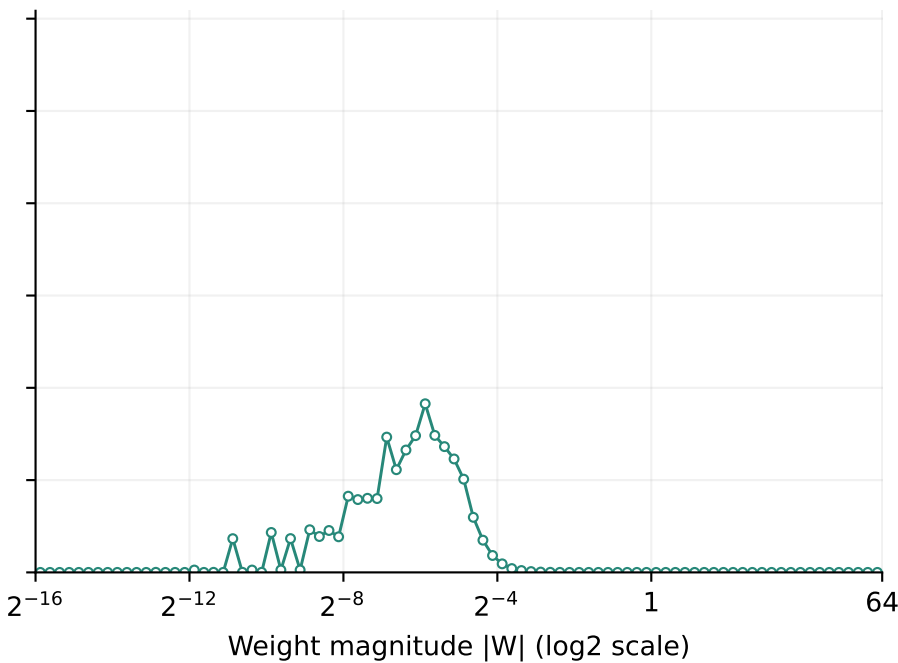
RMS 0.0204774
Max |W| 0.800781
Zero weights 0.0002%
FP16 reference
Nonzero mass outside x-axis 0.0663%

INT4 (Q4_HP1)



RMS 0.0206757
Max |W| 0.75
Zero weights 17.4142%
Relative L2 error 13.021%
SQNR 17.707 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0205358
Max |W| 0.804688
Zero weights 1.1014%
Relative L2 error 0.847%
SQNR 41.445 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

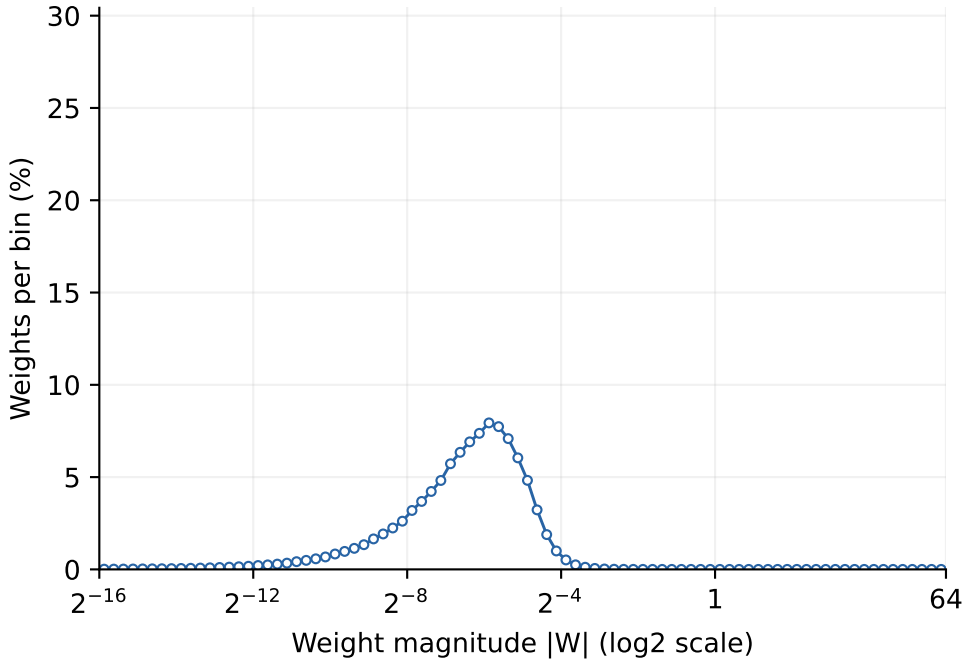
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

Llama-3.2-1B / Transformer block 15

7 matrix tensor(s) | 60,817,408 weights | FP16 vs PoTaI HP1

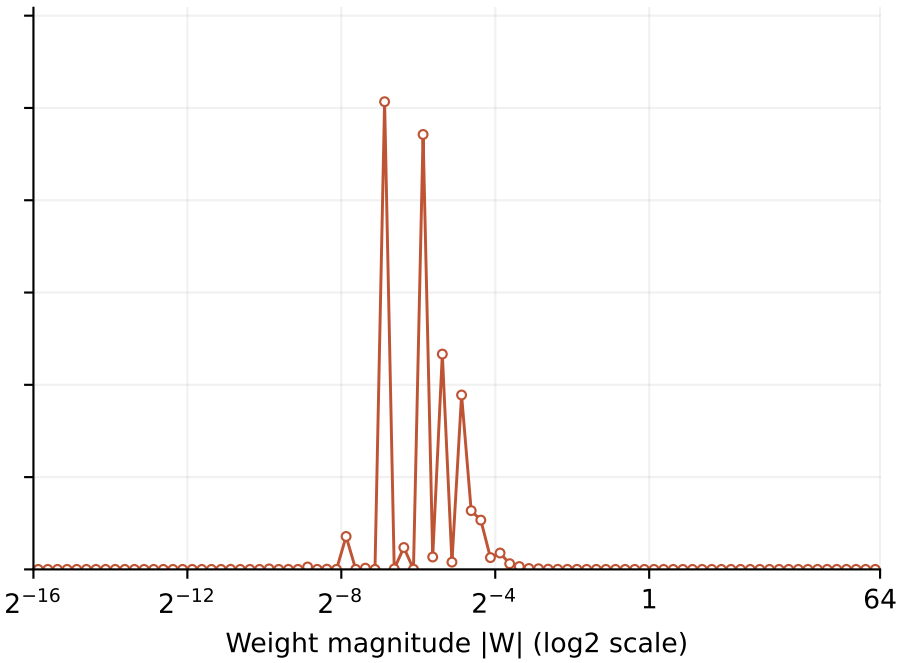
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



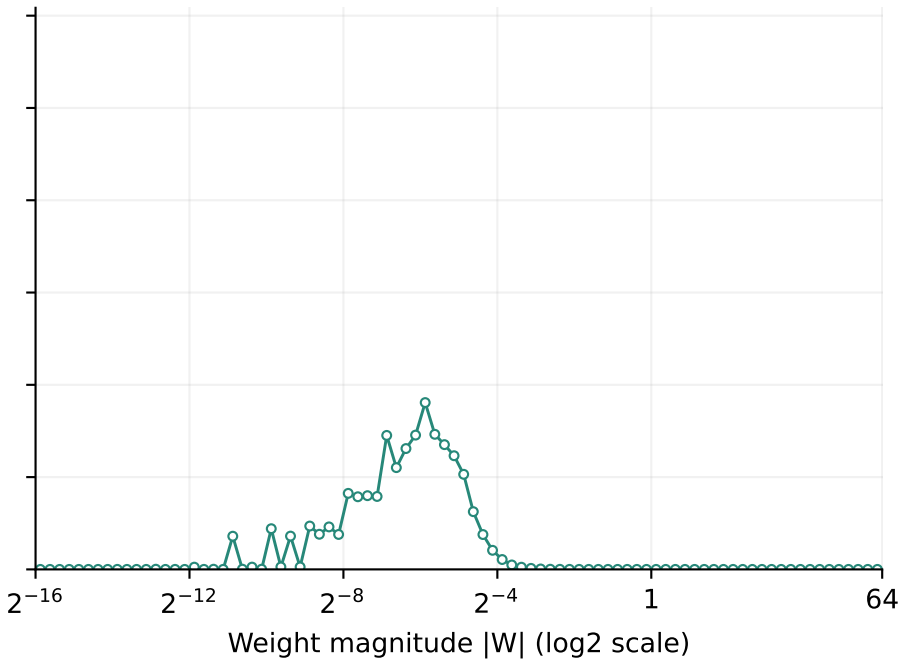
RMS 0.0210786
Max |W| 1.23438
Zero weights 0.0001%
FP16 reference
Nonzero mass outside x-axis 0.0671%

INT4 (Q4_HP1)



RMS 0.0212846
Max |W| 1.25
Zero weights 17.7084%
Relative L2 error 13.137%
SQNR 17.630 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.0211386
Max |W| 1.23438
Zero weights 1.1210%
Relative L2 error 0.853%
SQNR 41.378 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: ffn_down, ffn_gate, ffn_up, attn_k, attn_output, attn_q, attn_v.

Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.